Harsh Patnaik
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LinkedIn | GitHub
Location: Bengaluru, India

Education

PES University, Bengaluru

Bachelor of Technology (B.Tech) in Electronics and Communication Engineering

Expected Graduation: 2028

Technical Skills

Category	Technologies
Robotics & Simulation	ROS 2, Gazebo, OpenCV, PID Control, Linux
UAVs & Drone Tech	Betaflight, ELRS, CrazyF405HD FC, Radiomaster Ranger Nano, ESC Configurator
Embedded & IoT	ESP32, Arduino, Sensors (Gas, Ultrasonic, DHT11), I2C/SPI Protocols
Languages	Python, C, JavaScript, C++
Web Development	React, Next.js, Node.js, Express.js, React Native, HTML, CSS
Tools & Data	Git, MongoDB, Firebase, Pandas, Matplotlib, VS Code

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Key Projects & Experience

e-Yantra Robotics Competition (eYRC 2025) – Stage 2 Qualifier

Role: Robotics & Drone Engineer | Tech: ROS 2, Python, Betaflight, Gazebo, ELRS Hardware

- **Hardware Integration:** Set up the CrazyF405HD ELRS FC and Ranger Nano TX (ELRS link, channel mapping, arming, AUX, leveling) for stable drone operation.
- **Circuit Design:** Built three compact 555-timer latching indicator circuits for pesticide pickup and infected-plant identification, providing reliable trigger-latch behaviour with LED feedback.
- **Diagnostics & Troubleshooting:** Diagnosed critical motor direction mismatch and reconfigured ESC settings to establish correct Quad-X mixer logic, eliminating uncontrolled yaw spins.
- **Simulation & Control:** Designed ROS 2 nodes to simulate drone flight in Gazebo; implemented a PID-based stabilization algorithm achieving ±0.4m positional accuracy.
- **Computer Vision:** Developed image processing algorithms using OpenCV to detect ArUco markers and identify plant infections for precision agriculture tasks.

Recon Bot – Hackezee Hackathon 2025 (Runner-Up)

Role: Embedded Systems Lead | Tech: ESP32, Sensors, IoT

- Built a teleoperated reconnaissance robot designed for hazardous zone surveillance, integrating real-time sonar mapping and gas-level detection.
- Engineered a cloud-based dashboard to visualize sensor data instantly, reducing operational risk for rescue teams.
- Collaborated in a rapid development cycle to prototype and deploy the system within 8 hours.

Terminal-Based Adventure Runner Game

Role: C Programmer | Tech: C Language, Data Structures

- Built a procedural terminal game using state machines, dynamic memory allocation, and function-pointer-based attack handlers, featuring scalable enemy generation, dodge probability logic, and persistent player progression via binary file storage.

Other Projects

- **VigilAI – AI Misinformation Checker (Gemini API, Python, Web Scraping):** Built a prototype that classifies misinformation by detecting platform type (YouTube, Instagram, web), extracting content (text, audio, keyframes), and evaluating it through Gemini's multimodal tools using a Python backend and web-based frontend.
- **VoiceIt – Grievance Portal (MERN Stack):** Developed a full-stack grievance management platform with issue submission, categorization, tracking, and role-based admin tools; selected as a **Top 10 Finalist** at Codfusion Hackathon 2024.
- **Airline Booking System (Python, CustomTkinter, MySQL):** Developed a GUI-based application with basic credential validation, booking and cancellation workflows, admin tools, and MySQL-backed data management.
- **Machine Learning Models:** Utilized Scikit-learn for predictive modeling and data analysis.
- **COVID Tracker App (React Native):** Developed a mobile app for tracking real-time global pandemic statistics.

Research Interests

Autonomous Systems, UAV Dynamics & Control, Embedded Systems, IoT & Smart Devices, Robotics Simulation, Computer Vision.